05 Collisions and Momentum Conservation

Wednesday, December 13, 2023 1:03 AM

Chapter 8

Impulse (ΙΤνε) & Momentum & Collisions (Σύγκρουσα)

$$\sum \vec{F} = m\vec{a} = m\frac{d\vec{v}}{dt}$$

$$\underbrace{\sum \vec{F}\,dt}_{} = m\,d\vec{v}$$

$$\text{impulse} = \vec{J} = Ns =$$

$$\underbrace{\sum \vec{F}\,dt}_{\text{impulse}} = \underbrace{m\,d\vec{v}}_{\text{impulse}}$$

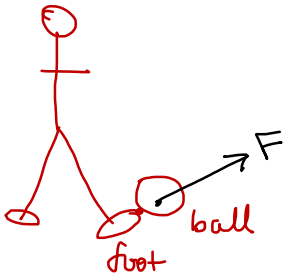
impulse ≡ change in 'momentum'

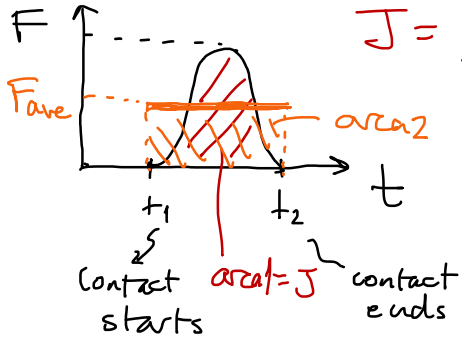
$$\vec{P} \equiv m\vec{v} \quad \text{(vector) momentum} = \left[kg\,\frac{m}{s}\right]$$

$$\underbrace{\sum \vec{F} = \frac{d\vec{P}}{dt}}_{} = \frac{d}{dt}(m\vec{v}) = \underbrace{\frac{dm}{dt}\vec{v}}_{\text{How rockets move in space.}} + m\frac{d\vec{v}}{dt} \equiv \text{Newto}$$

$$\sum \vec{F}\,dt = d\vec{P}$$

$$\underbrace{\sum \vec{F}\,\Delta t}_{\text{impulse}} = \underbrace{\Delta \vec{P}}_{\text{change in mom.}}$$





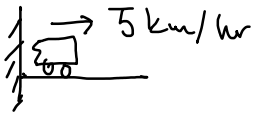
$$J = \underbrace{\int F\,dt}_{\text{area 1}} = \underbrace{F_{ave}\,\Delta t}_{} = \text{area 2}$$

$$\Delta t = t_2 - t_1$$

ex)

$v = 60$ km/hr

$m = 1000$ kg



5 km/hr $\rightarrow \hat{i}$

a car is colliding with a wall initial velocity is 60 km/hr to after collision velocity 5 km/hr The collision took 0.1 s.

$$\vec{F}_{ave}\,\Delta t = \vec{J} = \Delta\vec{P}$$

?

a) F_{ave} to car = ?

$$\vec{P} = m\vec{v}$$

$$= \vec{P}_f - \vec{P}_i \qquad \rightarrow +x$$

$v_i = 6$

$v_f =$

$$= m\vec{v}_f - m\vec{v}_i = 1000\left(\frac{5\hat{i}}{3.6} - \frac{60}{3.6}(-\hat{i})\right)$$

$$1000(1.4 + 16.7)$$

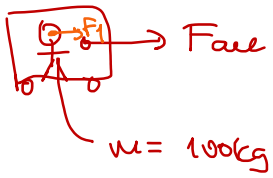
$$F_{ave} = \underline{\underline{181000\,N}}$$

$$\vec{F}_{ave}(0.1s) = 18100 \Rightarrow \vec{F}_{ave} =$$

$m = 100\,kg$

$mg \simeq 10^3 N$

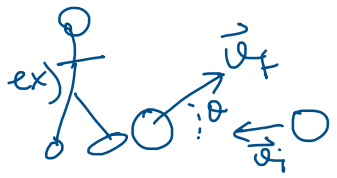
$F_{ave} = (180$ of $100\,kg$ person$)$



$F_{ave} = 181000 N$

$F_1 = 18100 N \Rightarrow$ (18 of 100kg person ...)

↳ why we buckle up!!



$v_i = 20\,m/s$

$v_f = 30\,m/s$

$\theta = 45°$

$m = 0.4\,kg$

$\Delta t = 0.01\,s$

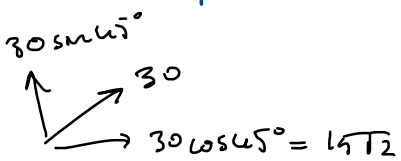
$\vec{J} = ?$

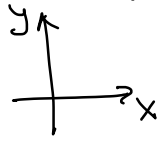
$\vec{F}_{ave} = ?$

$\vec{J} = \Delta\vec{P} = \vec{F}_a$

$m\vec{v}_f - m\vec{v}$

$\vec{P}_i =$





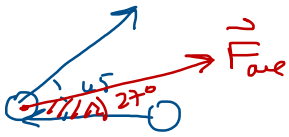
$$\vec{P}_f = 0.4\left(15\sqrt{2}\,\hat{i} + 15\sqrt{2}\,\hat{j}\right) kg\,\frac{m}{s}$$

$$\Delta\vec{P} = \vec{P}_f - \vec{P}_i = \left(6\sqrt{2}\,\hat{i} + 8\,\hat{i}\right) + 6\sqrt{2}\,\hat{j}\ kg\,\frac{m}{s} = 16.5\,\hat{i} -$$

$$\vec{F}_{ave} = \frac{\Delta\vec{P}}{\Delta t} = \left(1650\,\hat{i} + 850\,\hat{j}\right) N \qquad ; \left|\vec{F}_{ave}\right| = 1850$$

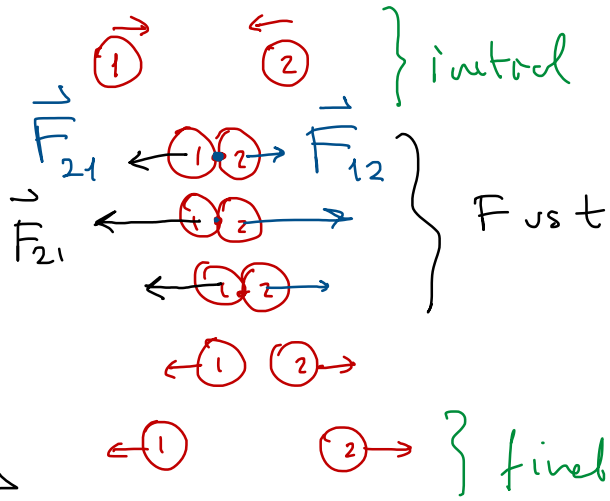
$\Delta t \to 0.01\,s$

$$\alpha = \tan^{-1}\left(\frac{850}{1650}\right) = 27$$



Momentum Conservation in Collisions

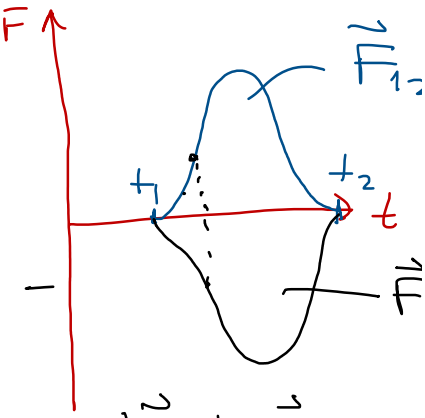
3rd law



initial

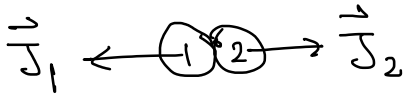
F vs t

final



$|\vec{F}_{12}| = |\vec{F}_{21}|$

$\vec{J} = \Delta\vec{P}$



area under curve

$\vec{F}_{12} = -\vec{F}_{21}$ $\qquad J_2 = -J_1$

$\underbrace{F_{12}\,\Delta t}_{} = -F_{21}\,\Delta t$

$\Delta\vec{P}_2 = -\Delta\vec{P}_1 \Rightarrow \Delta\vec{P}_2 + \Delta\vec{P}_1 = 0$

$\vec{P}_{2f} - \vec{P}_{2i} + \vec{P}_{1f} - \vec{P}_{1i} = 0$

initial = final

$\vec{P}_{1i} + \vec{P}_{2i} = \underbrace{\vec{P}_{1f} + \vec{P}_{2f}}$

$\underbrace{\Sigma\vec{P}_i = \Sigma\vec{P}_f}_{\text{momentum conservation}}$ $\qquad \Sigma\vec{P}_i = \Sigma\vec{P}_f$

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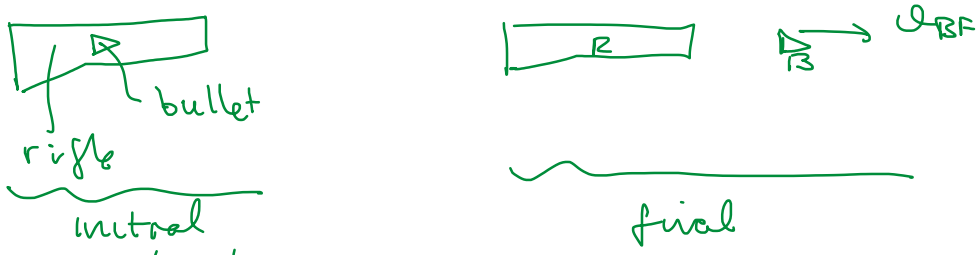
ex) Recoil of a rifle (tüfeğin geri tepmesi)



m_{Bullet}
m_{Rifle}
v_{BF}
a) v_{RF}

$v_{Bi} = 0$
$v_{Ri} = 0$

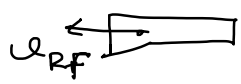
$\Sigma\vec{P}_i = \Sigma\vec{P}_f \quad \rightarrow +x$

$\vec{P}_{Ri} + \vec{P}_{Bi}$

$0 + 0 = m_B\vec{v}_{BF} + m_R\vec{v}_{RF}$

$0 = (0.005)(300\hat{i}) + 3\,\vec{v}_{RF}$

$\vec{v}_{RF} = \dfrac{-1.5\hat{i}}{3} = \underline{\underline{-0.5\ m/s\ \hat{i}}}$



b) what's the change in Kinetic Energy of Rifle and Bullet?

$K_{Bi} = 0 \qquad K_{Ri} = 0$

$\frac{1}{2} m_B v_{BF}^2 = K_{BF} = 22$

$\frac{1}{2}(0.005)300^2$

$\Delta K_B = K_{BF} - K_{Bi} = 225\,J$

change

$\frac{1}{2} m_R v_{RF}^2 = 0.375\,J$

$\Delta K_R = \underbrace{0.375\,J}_{?!}$

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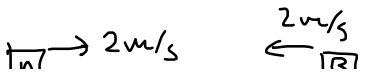
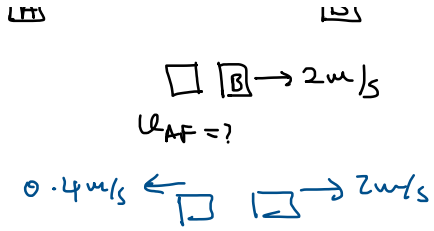
ex



$M_R = 0.3\,kg$ $\qquad \Sigma\vec{P}_i = \Sigma\vec{P}_f$



$m_A = 0.5\,kg$

$\longrightarrow +x$

$m_A \vec{v}_{Ai} + m_B \vec{v}_{Bi} = m_B v$

$(0.5)\,2\hat{i} + (0.3)(-2\hat{i}) =$

$\dfrac{(0.4 - 0.6)\hat{i}}{0.5} =$

$v_{AF} = -0.4$

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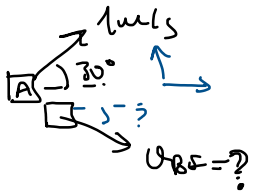
ex)

$m_A = 20\,kg$

$m_B = 12\,kg$

A → 2m/s

B, $v_{Bi} = 0$



$$\sum \vec{P}_i = \sum \vec{P}_f$$

$$20(2\hat{i}) + 12(0) = 20(1\cos 30\,\hat{i} + 1\sin 30\,\hat{j})$$

$\hat{i} \Rightarrow$ $40\hat{i} = 20\cos 30\,\hat{i} + 12 v_{Bx}\hat{i}$

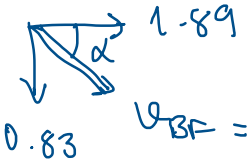
$\hat{j} \Rightarrow$ $0\hat{j} = 20\sin 30\,\hat{j} + 12\, v_{By}\hat{j}$

$v_{By} \uparrow$

$v_{By} = \dfrac{-10}{12}\hat{j}\ m/s \Rightarrow$ $\downarrow v_{By}$

$v_{By} = -$

$\dfrac{40 - 17.3}{12} = v_{Bx} = 1.89\,\hat{i}\ m/s$



$\alpha = \tan^{-1}$

$v_{BF} = \sqrt{1.89^2 + 0.83^2} = 2.06\ m/s$

COLLISIONS ⇒ momentum is CONSERVED.

ELASTIC

Kinetic Energy is CONSERVED

$\sum K_i = \sum K_f$

INELASTIC

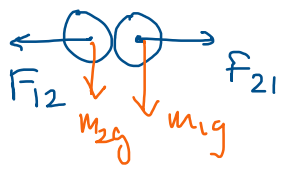
K.E is NOT CONSERVED.

$\sum K_i \neq \sum K_f$

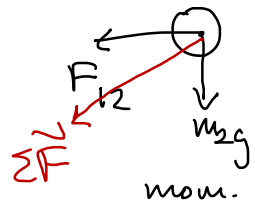
most $\sum K_i > \sum K_f$

except explosion $\sum K_i < \sum K_f$

$\Sigma \vec{P}_i \neq \Sigma \vec{P}_f$ not Geip conserved



$\underbrace{\Sigma \vec{F}}_{\text{all the forces}} = \frac{d\vec{P}}{dt}$



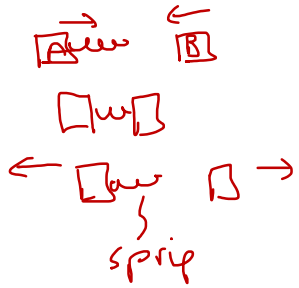
Σ

mom. is n
cons

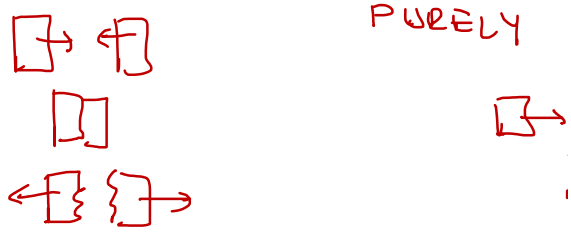
$\underbrace{m_2 g \ll F_{12}}$ i

(mom is cons

ELASTIC



INELASTIC COLL. → PURELY



ex) $\circ_i$ A → 2 m/s 2 m/s ← B

0.5 kg 0.3 kg

what's the % of energy lost after the collis

after collision A & B move

f AB v_F

$\Sigma \vec{P}_i = \Sigma \vec{P}_f \longrightarrow +x$

$0.5(2\hat{i}) + 0.3(-2\hat{i}) = (0.8)\,\vec{v}_F$

$+\frac{0.4}{0.8}\hat{i} = \vec{v}_F = 0.5\,\hat{i}$ m/s

$\Sigma K_i = \frac{1}{2}$

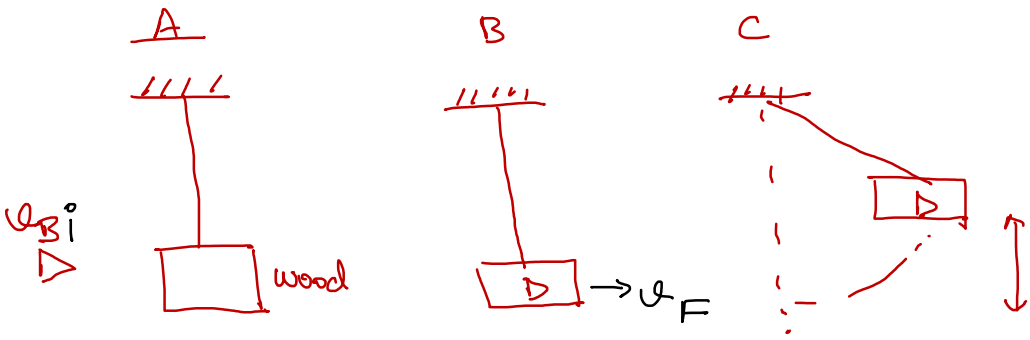
$=$

$\Sigma K_f =$

1.6 J → 0.1 J

$\frac{|\Delta K|}{K_i} = \frac{|0.1 - 1.6|}{1.6} = 0.9$

BALLISTIC PENDULUM



(I) A → B collision

$$\sum \vec{P}_A = \sum \vec{P}_B$$

(II) $B \to C$ energy conservation

m_B, m_w ; $v_{wi} = 0$ $\qquad \sum \vec{P}_i = \sum \vec{P}_f$

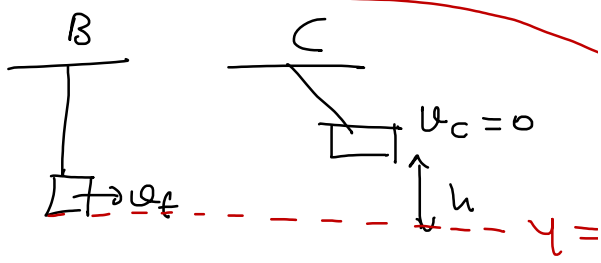
$$m_B \vec{v}_{Bi} + m_w \underbrace{\vec{v}_{wi}}_{0} = (m_B + m_w)\,\bar{v}$$

$$v_F = \frac{m_B\, v_{Bi}}{(m_B + m_w)}$$

II) $E_B = E_C$

$$K_B + \cancel{U_B} = \cancel{K_C} + U_C$$

$y_B = 0$ $\qquad 0$ $\quad v_C = 0$



$$\frac{1}{2}(m_B + \cancel{m_w})\, v_F^2 = (m_B + \cancel{m_w})\, g h \Rightarrow v_F = \sqrt{}$$

$$\sqrt{2gh} = \frac{m_B}{(m_B + m_w)}\, v_{Bi} \Rightarrow v_{Bi} = \frac{m_B + m_w}{m_B}$$

if $m_B = 5gr$
$m_w = 2kg$
$h = 3cm$

$$v_{Bi} = \frac{2.005}{0.005}\sqrt{2(9.8)(0.03)} = 3c$$

$$v_F = \sqrt{2gh} = 0.76\ m/s$$

How much energy is lost?

$$\sum K_i = \frac{1}{2}(0.005)(307)^2 = 236\,J$$

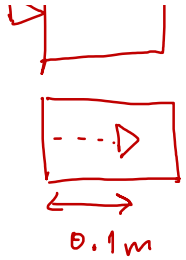
$$\sum K_F = \frac{1}{2}(2.005)\,0.76^2 = (2.005)(9$$

$$= 0.6\,J$$

$$\text{Lost} = \frac{236 - 0.6}{236} = 0.997 \quad 99.7\%$$ is

where does the energy go?

$W_{other} + E_i = E_f$



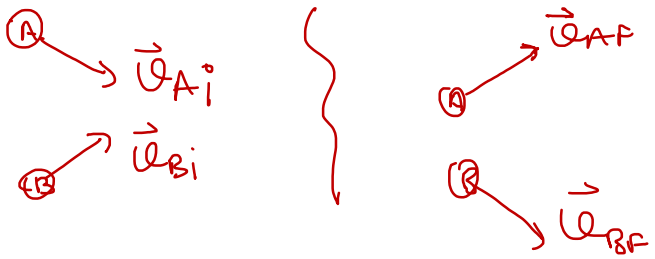
average frictional force on the bullet?

$W_f + 236\,J = 0.6\,J$

$W_f = -235.4\,J$

$f =$

ELASTIC COLLISION



m_A v_{Ai}
m_B v_{Bi}

6 te

$$m_A\vec{v}_{Ai} + m_B\vec{v}_{Bi} = m_A\vec{v}_{AF} + m_B\vec{v}_{BF} \qquad \Sigma\vec{P}_i$$

$$\frac{1}{2}m_A v_{Ai}^2 + \frac{1}{2}m_B v_{Bi}^2 = \frac{1}{2}m_A v_{AF}^2 + \frac{1}{2}m_B v_{BF}^2 \qquad \Sigma$$

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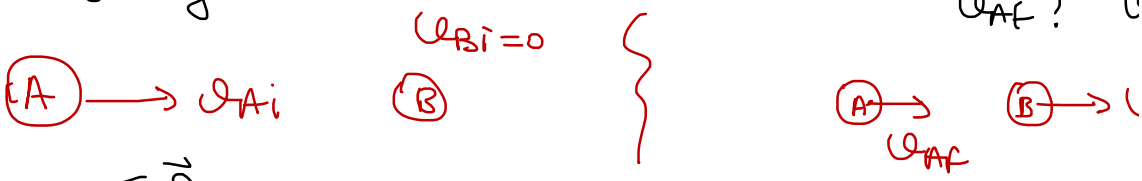
1 dimensional Elastic Collision.

m_A m_B v_{Ai} v_{Bi} ⇒ given

v_{AF}? v



$$\Sigma\vec{P}_i = \Sigma\vec{P}_f$$

$$m_A v_{Ai} + 0 = m_A v_{AF} + m_B v_{BF}$$

$$\Sigma E_i = \Sigma E_f$$

$$\cancel{\tfrac{1}{2}}m_A v_{Ai}^2 + 0 = \cancel{\tfrac{1}{2}}m_A v_{Af}^2 + \cancel{\tfrac{1}{2}}m_B v_{Bf}^2$$

$$m_B v_{Bf}^2 = m_A\left(v_{Ai}^2 - v_{Af}^2\right) \quad — (1)$$

$$m_B v_{BF} = m_A\left(v_{Ai} - v_{AF}\right) \quad — (2)$$

(1)

$$\boxed{v_{BF} = v_{Ai} + v_{AF}} \rightarrow \text{into } (2)$$

$$m_B(v_{Ai} + v_{AF}) = m_A(v_{Ai} - v_{AF})$$

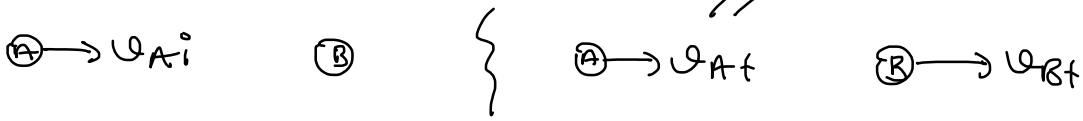
$$m_B v_{AF} + m_A v_{AF} = m_A v_{Ai} - m_B v_{Ai}$$

$$\boxed{v_{AF} = \frac{m_A - m_B}{m_A + m_B} v_{Ai}}$$

$$v_{BF} = v_{Ai} + v_{Ai}\left(\frac{m_A - m_B}{m_A + m_B}\right)$$

$$v_{Ai}\left(\frac{m_A + \cancel{m_B} + m_A - \cancel{m_B}}{m_A + m_B}\right)$$

$$\boxed{v_{BF} = v_{Ai}\frac{2m_A}{m_A + m_B}} \qquad v_{AF} = \frac{m_A - m_B}{m_A + m_B}$$



case I) $m_A = m_B = m$

$$v_{AF} = \frac{m - m}{2m} v_{Ai} = 0 \qquad v_{BF} = v_{Ai}\frac{2m}{m+m} = v$$

$$v_{BF} = v_{Ai}\frac{2m_A}{m_A + m_B} \qquad v_{AF} = \frac{m_A - m_B}{m_A + m_B} v_{Ai}$$

case II) $m_A \gg m_B$

$$\frac{m_A}{m_B} = 1000$$

$$v_{AF} = \frac{1000 - 1}{1000 + 1} v \simeq v$$

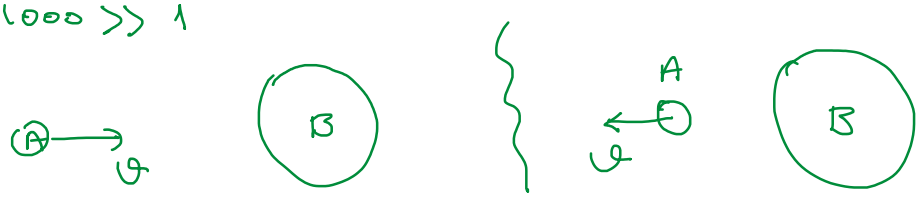
$$v_{BF} = v\frac{2(1000)}{1000 + 1} \simeq 2v$$

$$v_{BF} = v_{Ai} \frac{2m_A}{m_A+m_B} \qquad v_{AF} = \frac{m_A - m_B}{m_A+m_B} v_{Ai}$$

Case III)

$m_B \gg m_A$

$1000 \gg 1$



$$v_{Bf} = v \frac{2(1)}{1001} = 0{,}002\, v$$

$$v_{AF} = v \frac{1-1000}{1001} \simeq -v$$

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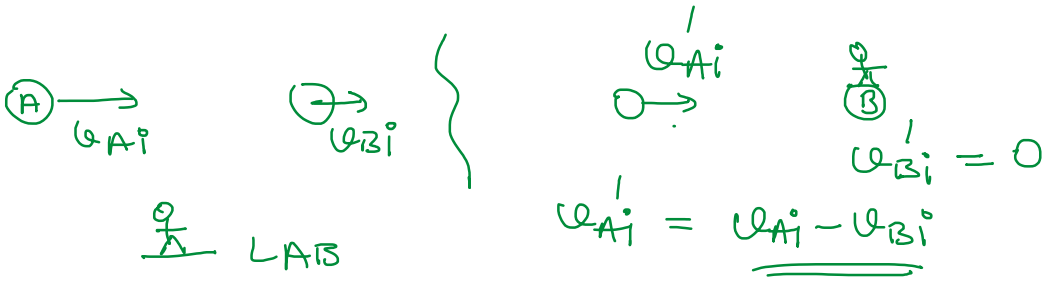
1 dim elastic collisions



$$v'_{Ai} = v_{Ai} - v_{Bi}$$

$$v'_{BF} = v'_{Ai} \frac{2m_A}{m_A+m_B} \qquad v'_{AF} = \frac{m_A - m_B}{m_A+m_B} v'_{Ai}$$

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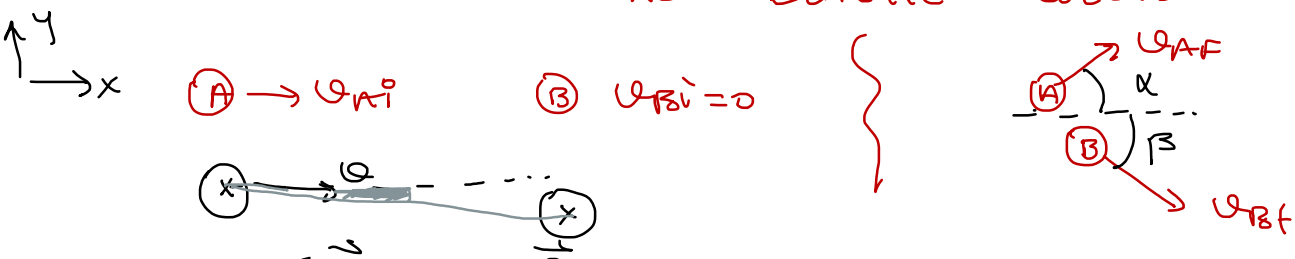
2 DIMENSIONAL ELASTIC COLLISION



$$\sum \vec{P}_i = \sum \vec{P}_f$$

$$m_A v_{Ai} \hat{i} + 0 = m_A v_{AF} \cos\alpha\, \hat{i} + m_A v_{AF} \sin\alpha\, \hat{j} + m_B v_{BF} \cos\beta\, \hat{i} - m_B v_{BF} \sin\beta\, \hat{j}$$

$\hat{i} \Rightarrow \quad m_A v_{Ai} = m_A v_{AF} \cos\alpha + m_B v_{BF} \cos\beta \quad$ — (1)

$\hat{j} \Rightarrow \quad 0 = m_A v_{AF} \sin\alpha - m_B v_{BF} \sin\beta \quad$ — (2)

$\sum K_i = \sum K_f$

$$\cancel{\tfrac{1}{2}} m_A v_{Ai}^2 + 0 = \cancel{\tfrac{1}{2}} m_A v_{Af}^2 + \cancel{\tfrac{1}{2}} m_B v_{Bf}^2$$

$$\left\{\begin{array}{lll} m_A = 0.5\ kg & v_{Ai} = 4\ m/s & v_{Bi} = 0 \\ m_B = 0.3\ kg & v_{Af} = 2\ m/s & \end{array}\right\} \quad \alpha,\ \beta,\ v_{Bf}\ ?$$

$$(0.5)(4) = (0.5)(2)\cos\alpha + (0.3)\, v_{Bf} \cos\beta$$

$$0 = (0.5)(2)\sin\alpha - (0.3)\, v_{Bf} \sin\beta$$

$$(0.5)\, 4^2 = (0.5)\, 2^2 + (0.3)\, v_{Bf}^2$$

(3) $$8 = 2 + 0.3\, v_{Bf}^2 \Rightarrow v_{Bf} = \sqrt{20} = \underline{\underline{4.47\ m}}$$

(1) $$2 = \cos\alpha + 0.3\sqrt{20}\cos\beta$$

(2) $$0 = \sin\alpha - 0.3\sqrt{20}\sin\beta$$

$$\cos\beta = \frac{2 - \cos\alpha}{0.3\sqrt{20}} \Rightarrow \cos^2\beta = \frac{1}{0.3^2(20)}\left[4 + \cos^2\alpha\right.$$

$$\sin\beta = \frac{\sin\alpha}{0.3\sqrt{20}} \Rightarrow \sin^2\beta = \frac{1}{0.3^2(20)}\sin^2\alpha$$

$$+$$

$$1 = \frac{1}{0.3^2(20)}\left[4 + \right.$$

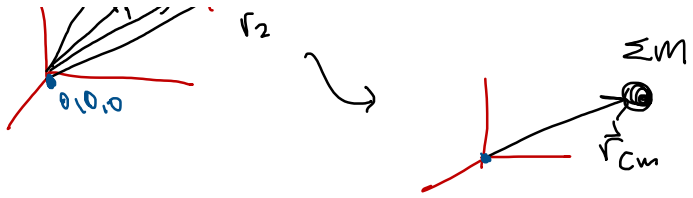
$$\cos\alpha = 0.8 \Rightarrow$$

$$\cos\beta = \frac{2 - (0.8)}{0.3\sqrt{20}} = 0.89$$

$$\beta = \cos^{-1}(0.89) = 26.6 \simeq \underline{\underline{27^\circ}}$$

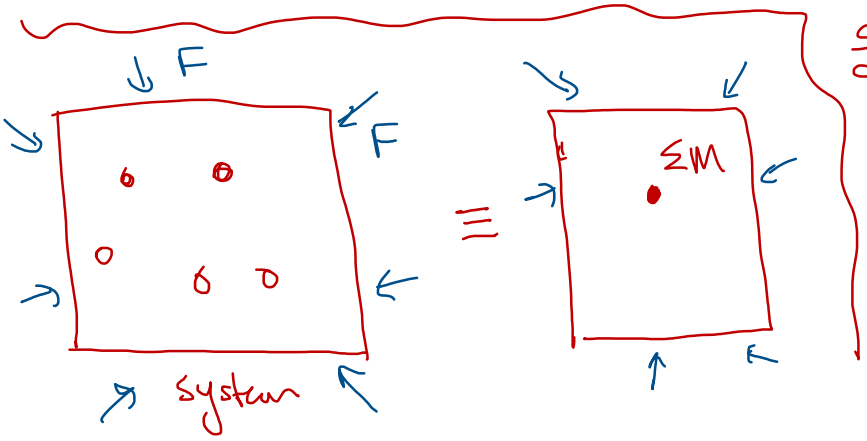
Center of Mass

$$\vec{r}_{cm} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + m_3\vec{r}_3 + \cdots + m_N\vec{r}}{}$$



$m_1 + m_2 + m_3 + \cdots + m_N$

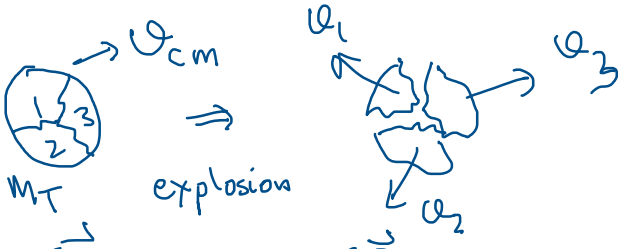
$$\vec{r}_{cm} = X_{cm}\hat{i} + Y_{cm}$$

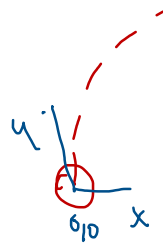


$$\frac{d}{dt}\downarrow \quad \vec{v}_{cm} = \frac{\sum m_i \frac{d\vec{r}_i}{dt}}{\sum m_i}$$

$$\frac{d}{dt}\downarrow \quad \vec{a}_{cm} = \frac{\sum_{i=1}^{N} m_i \vec{a}_i}{\sum m_i}$$

$\Sigma F_{\text{external on to system}} = 0 \Rightarrow$ momentum conservation





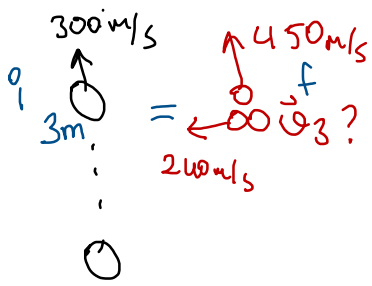
$$\Sigma \vec{P}_i = \Sigma \vec{P}_f$$

$$M_T \vec{v}_{cm} = m_1\vec{v}_1 + m_2\vec{v}_2 + m_3\vec{v}_3$$

ex) An object is thrown vertically ; at 1000 m above the grou

300 m/s $\hat{j}$; at this moment object splits into 3 equal p

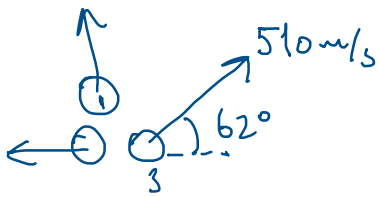
1st part 450 m/s $\hat{j}$; 2nd part 240 m/s toward



$$\Sigma \vec{P}_i = \Sigma \vec{P}_f$$

$$3\cancel{m}(300\hat{j}) = \cancel{m}\,450$$

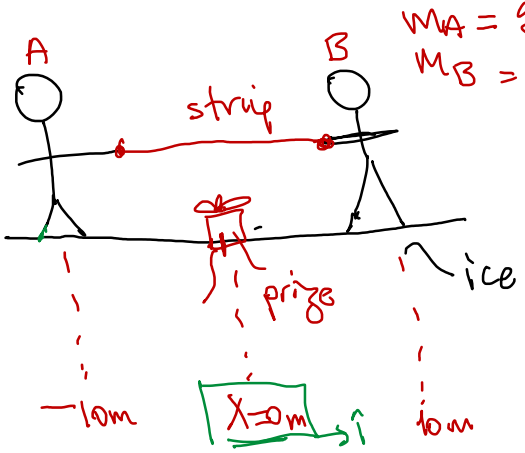
$$\vec{v}_3 = (240\hat{i}$$



$\theta = \tan$

$v_3 = \sqrt{240^2}$

ex) Tug of war (ip, natural venue)

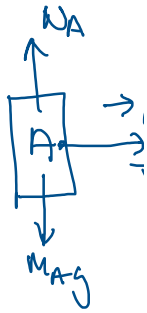


$m_A = 90\,kg$
$m_B = 60\,kg$

whoever gets to the p

who?

B reaches the midpoint first!!



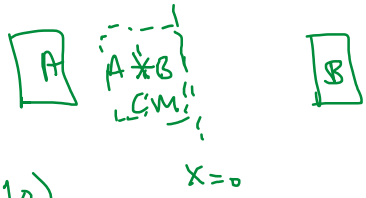
$\Sigma F =$

$\frac{I}{m_A}$

CM (center of mass)

$$X_{cm} = \frac{m_A x_A + m_B x_B}{m_A + m_B} \Rightarrow X_{cm} = \frac{90(-10) + 60(10)}{150} = \frac{-300}{150}$$

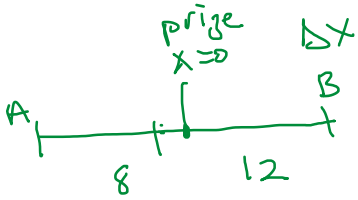
B will go from $x = 10m$, $x = -2m$



A will n
max d
-1
A

$\Delta x = -2 - (10)$

$\Delta x = -12m$



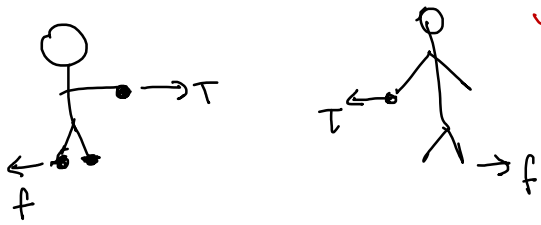
↳ B will win!!

$v_{cm} = \frac{m_A v_A}{m}$

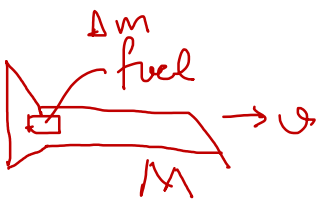
$\Downarrow$

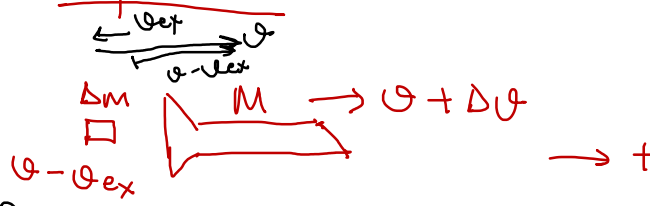
0 no initial vel.

⇒ Tug of war with friction



Rocket Propulsion





* v_{ex} = exhaust velocity

$$\Sigma \vec{P}_i = \Sigma \vec{P}_f$$

$$(M + \Delta m)\, v = \Delta m\,(v - v_{ex}) + M(v + \Delta v)$$

$$Mv + \Delta m v = \Delta m v - \Delta m v_{ex} + Mv + \Delta m v_{ex} = M \Delta v$$

farklı

Total mass = Rocket metal + fuel

$$1000\,kg = 300\,kg + 700\,kg$$
$$999 = 300 + 699$$
$$900 = 300 + 600$$

decrease

$-\Delta M$

$M_f - M_i$

$999 - 1000 = -1$

exhausted fuel

0

1

100

increase $+$

$\Delta m = 1 - 0 = 1$

$\Delta m =$

$M_i =$

?!

$$\Delta m v_{ex} = M \Delta v$$

$$-\Delta M v_{ex} = M \Delta v \Rightarrow \boxed{-dM v_{ex} = M dv}$$

$-$

$$\int_{i}^{f} \frac{dM}{M} = \int \frac{-1}{v_{ex}} dv$$

$$\ln M \Big|_{M_i}^{M_f} = -\frac{1}{v_{ex}} \Delta v \Big|_{v_i}^{v_f} \rightarrow \underbrace{\ln M_f - \ln M_i}_{(-)} = -$$

$$v_{ex} \ln \frac{M_f}{M_i} = -v_f + v_i \Rightarrow v_f = v_i - v_{ex} \ln \frac{M_f}{M_i} =$$

$$-\frac{dM}{dt} v_{ex} = M \frac{dv}{dt}$$

M = total mass = (fuel + other)

$$-v_{ex}\left(\frac{dM}{dt}\right) = M \frac{dv}{dt} = Ma = \text{Force} \Rightarrow -v_{ex} \frac{dM}{dt} = F$$

$(-)\ (-) = +$

$$\frac{d\vec{P}}{dt} = \vec{F} = \frac{d}{dt}(m\vec{v}) = \frac{dm}{dt}\vec{v} + m\frac{d\vec{v}}{dt} = \vec{F}$$

ex) A rocket ejects fuel in the 1st second; it ejects 1

with a speed of 2400 m/s. what's acc. of rocket?

$$-\frac{dM}{dt} v_{ex} = M\frac{dv}{dt} = Ma$$

$$\frac{120}{120}M \rightarrow \frac{119M}{120}$$

$$-\left(-\frac{1}{120}\right)\cancel{M}(2400) = \cancel{M}a = \frac{2400}{120} = 20\ m/s^2 \simeq 2g$$

b) If $v_i = 0$ m/s ; $\frac{3}{4}$ th of the Mass of rocket fuel ⇒ this

$v_f = ?$

$$v_f = v_i + v_{ex}\ln\frac{M_i}{M_f} = 0 + 2400$$

c) If M = 1000 kg fuel is ejected in 90 seconds, what's on the rocket?

$$F = v_{ex}\frac{dM}{dt} = 2400\left(\frac{\frac{3M}{4}}{90}\right) =$$

---

ex If water ejected at $\frac{3600\ L}{minute}$ from a fire hose, is 600 N ; $v_{ex} = ?$ of water.

1 Lt = 1 kg 1 min = 60 s

$$\frac{\Delta M}{\Delta t} = \frac{3600}{60}$$

$$F = v_{ex}\frac{\Delta M}{\Delta t}$$

$$v_{ex} = \frac{F}{\Delta M/\Delta t}$$